Afnan Mohammad Hafiz

ID : 23241004

Sec : 19

CSE340

# Ans. to the ques. no. 1

## (a)

$$\underbrace{0000\ 0000\ 1111}_{\text{Immediate}}\ \underbrace{0010\ 1}_{rs1}\underbrace{001}_{funct3}\ \underbrace{1010\ 1}_{rd}\underbrace{001\ 0011}_{opcode}$$

Opcode = 0010011 which is I-type

This is denoted by LD, LW, LH, LB, ADDI, SLLI, SRLI, ORI

Here,

funct3 = SLLI

rs1 = $X_5$

rd = $X_{21}$

Immediate = 16

∴ The RISC-V code – SLLI $X_{21}$, $X_5$, 15

## (b)

0x089739B3

$\underbrace{0000\ 100}_{funct7}\underbrace{0\ 1001}_{rs2}\ \underbrace{0111\ 0}_{rs1}\underbrace{011}_{funct3}\ \underbrace{1001\ 1}_{rd}\underbrace{011\ 0011}_{opcode}$

Opcode = 0110011 which is of R-Type

funct 3 = OR

rd = $X_{19}$

rs1 = $X_{14}$

rs 2 = $X_9$

∴ The RISC-V code is — OR $X_{19}$, $X_{14}$, $X_9$.

## Ans. to the ques. no. 2

```
Loop:

    addi X5, X11, 13

    addi X6, X13, -4

    blt  X5, X6, Exit

    bge  X12, X11, Elif

    bge  X12, X13, Elif

    add  X11, X12, X6

    beq  X0, X0, loop
```

~~Elif:~~

```
Elif:

    addi X7, X0, 5

    bne  X11, X7, Else

    addi X13, X13, 10

    beq  X10, X0, loop
```

Else:

addi $X_{14}$, $X_{14}$, 10

beq $X_0$, $X_0$, loop

Exit

<u>Ans. to the ques. no. 3</u>

Jalr and Store falls under I-type and S-type format respectively.

The funct3 field specifies the type and size of the data being loaded from the memory. It tells the processor how many bytes to load and how to interpret them. Load belongs to I-type format and their types are –

lb = Load byte (8 bit)
lh = Load half word (16 bit)
lw = Load word (32 bit)
ld = Load double word (64 bit)

## Ans. to the ques. no. 4

### (a)

beq $X_0, X_0$, Loop is a SB type instruction.

$rs1 = X_0 = 00000$

$rs2 = X_0 = 00000$

funct 3 = 011

Opcode = $(103)_{10}$ = 1100111

Jumping offset = 7028 - 7056

= -28

$\therefore (28)_{10}$ = 0000 0001 1100

1bit extension = 00000 0001 1100

11111 1110 0011

+1

$\therefore (-28)_{10}$ = 11111 1110 0100

12 11 [10:5] [4:1] Discard

Loop:

\# 7028 Slli $X_{10}, X_{22}, 3$

\# 7032 Add $X_{10}, X_{10}, X_{25}$

\# 7036 ~~Addi~~ ld $X_9, 0(X_{10})$

\# 7040 Addi $X_{17}, X_0, -7$

\# 7044 Bne $X_9, X_{26}$, Exit

\# 7048 Addi $X_{22}, X_{22}, 1$

\# 7052 Sd $X_{10}, 16(X_{22})$

\# 7056 Beq $X_0, X_0$, Loop

Exit

\# 7060 Empty line

| Imm | Imm | rs2 | rs1 | funct3 | Imm | 11th | Opcode |
|---|---|---|---|---|---|---|---|
| 1 | 111111 | 00000 | 00000 | 011 | 0010 | 1 | 1100111 |

## (b)

Bne $X_9$, $X_{26}$, Exit

Jumping offset = 7060 - 7044

= 16

$rs1 = X_9 = 01001$

$rs2 = X_{26} = 11010$

$(16)_{10} = 0000\ 00010000$

1 bit extension = 0 0000 0001 0000

12 11 [10:5] [4:1] Discard

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 0 | 000000 | 11010 | 01001 | 010 | 1000 | 0 | 1100111 |